Turn legal expertise into reliable software.

Learn the decisions.
Make the rules explicit.
Check the result with the team.

The lawyer decides legal applicability.

LAWYERS

Approve the reasoning

RULES

Assemble and explain

AI REVIEW

Suggest checks and sources

Basis: original internship posting supplied by the candidate. Proposed understanding, not an internal-system assessment.

My earlier work in legal question answering

North South University · Bangladeshi statutes · early research contribution

1

Review the problem

Literature review and problem identification

2

Organise sources

Initial design: keep sections and their Act names together

3

Move to LLM work

Another sub-team continued the legal research

Earlier research asked: “Where is the evidence?”
This recent demo also asks: “When should the paragraph be used?”

Earlier responsibilities based on candidate account. The recent prototype is separate from the published implementation.

Prepare the sources and paragraph rules

Cell 1 identifies the evidence. Cell 2 defines when each prepared paragraph may be used.

1 · LEGAL SOURCES

10 sample summaries
Act + section + source ID

→

2 · SOURCE DETAILS

Status + simulated judgment notes

→

3 · MODULE RULES

7 paragraphs: conditions, dependencies and conflicts

→

4 · VERSIONED INPUTS

Order + source links
Knowledge base v1.2.0

EXAMPLE · RECORDS MODULE M5

Include only if records are available AND procedural module M3 is selected.
Place at order 50; link the paragraph to source C167.

Teaching data: illustrative summaries and fictional updates; no automatic legal-source downloads.

Build the draft — and explain every selection

Deterministic means the same facts and rule version produce the same result.

1 · VALIDATE

Recognised fields; valid text, forum and Booleans

→

2 · EVALUATE

Required facts match? Prerequisites selected?

→

3 · SELECT + TRACE

Include or skip each module; record why

→

4 · CHECK CONFLICTS

Both exclusive routes? Stop with an error

→

5 · ASSEMBLE

Fill selected templates in order; return hashes

SAMPLE OUTPUT

Jurisdiction → Facts → Procedure → Records → Prayer
Substantive grounds and bail are skipped because their flags are false.

Before assembly: reject duplicate module IDs, missing or incorrectly ordered dependencies, and invalid conflict references. Return the draft with source IDs, version and audit trace.

Retrieve evidence; separately check the draft

Two paths: similarity search finds passages; explicit links and dates drive document warnings.

A · RETRIEVAL AND QUERY DECOMPOSITION

1 · SEARCH

TF-IDF + cosine
Compare whole query

2 · APPLY GATE

Accept score ≥ 0.80
Example: 0.788 fails

3 · SPLIT IF NEEDED

No hit + multiple parts:
split at ; or “and”

4 · SEARCH PARTS

Same 0.80 gate
Merge + deduplicate

Accepted whole-query hits return directly. Failed parts are unresolved. Exact-text example returns C54 + P420.

B · DRAFT REVIEW

CHECK RESULT

Module IDs + KB
fingerprint match

CHECK CITATIONS

Missing required or
unknown/unlinked IDs?

CHECK UPDATES

Linked judgment note
newer than KB date?

RETURN FINDINGS

Sample: 3 update alerts
M3/C54, M3/C167,
M5/C167

0.80 is uncalibrated, not 80% accuracy. Updates are simulated. No generative LLM is used in this prototype.

Regression tests protect expected behavior

Each check compares actual behavior with a written expectation; an expected error can mean a test passes.

1 · REPEAT

Same case × 25
Identical text + full result

2 · COMPARE

4 synthetic cases
Expected modules + citations

3 · CHALLENGE

Both exclusive routes on
Consistent ConflictError

4 · UPDATE

Inject C167 judgment note
Warnings on M3 and M5

RUN ASSERTIONS

Compare results with expected behavior

REPORT PASS / FAIL

All 4 groups passed
Extra citation/search checks

STOP ON FAILURE

Final assertion fails if any test group fails

Fixtures are fictional, not lawyer-approved historical cases. Passing tests does not establish legal correctness.

PROPOSED NEXT STEPS

How I would keep it reliable in real use

Start with one approved workflow, then expand only after review.

BEFORE THE PILOT	**Use approved material**	Verified full sources, agreed rules and authorised example cases.
WITH EACH CHANGE	**Version, test and review**	Rerun agreed cases; explain intended and unexpected differences.
BEFORE EXPANDING	**Evaluate the weak points**	Measure retrieval and citation support; keep legal decisions with reviewers.

Which workflow would be the most useful starting point for this internship?

Proposed improvements based on the brief and demo limitations, not verified shortcomings in the existing company system.